http://www.mjpruonline.com

Paper Code : 21304

F–404

B. C. A. (First Semester)
EXAMINATION, 2016-17
(New Course)
Paper No. BCA–104–N

MATHEMATICS-I

Time : Three Hours] [*Maximum Marks : 70*

Note : Attempt any *five* questions. All questions carry equal marks. Symbols used have their usual meanings.

1. (a) Show that $\lim_{x \to 0} \left(\frac{\tan x}{x} \right)^{\frac{1}{x^3}}$ does not exist.

   (b) Discuss the types of discontinuities.

2. (a) A function $f(x)$ is defined as :

$$f(x) = \begin{cases} \frac{x^2}{a} - a & : \ x < a \\ 0 & : \ x = a \\ a - \frac{a^2}{x} & : \ x > a \end{cases}$$

   Show that the function $f(x)$ is continuous at $x = a$.

   (b) Find nth derivative of $\tan^{-1} \frac{x}{a}$ with respect to x.

http://www.mjpruonline.com

http://www.mjpruonline.com

3. (a) If $x = \sin \left\{ \frac{(\log y)}{a} \right\}$, prove that :

$$(1 - x^2) y_{n+2} - (2n + 1) x y_{n+1} - (n^2 + a^2) y_n = 0$$

   (b) If $u = e^{xyz}$ show that :

$$\frac{\partial^3 u}{\partial x \, \partial y \, \partial z} = e^{xyz} \{ 1 + 3xyz + x^2 y^2 z^2 \}$$

4. (a) If $x = a \cosh \xi \cos \eta$,

$y = a \sinh \xi \sin \eta$

   prove that :

$$\frac{\partial (x, y)}{\partial (\xi, \eta)} = \frac{a^2}{2} (\cosh 2\xi - \cos 2\eta)$$

   (b) If $I_n = \int_0^{\frac{\pi}{3}} \tan^n x \, dx$, show that :

$$(n - 1) (I_n + I_{n-2}) = \left(\sqrt{3} \right)^{n-1}$$

5. (a) Evaluate $\int_0^{\frac{\pi}{2}} \log \sin x \, dx$

   (b) Find the entire area between the curve $y^2 (2a - x) = x^3$ and its asymptote.

6. (a) Solve :

$$\frac{x \, dx + y \, dy}{x \, dy - y \, dx} = \sqrt{\frac{a^2 - x^2 - y^2}{x^2 + y^2}}$$

   (b) Solve :

$$(D^2 + D + 1) y = \sin 2x$$

http://www.mjpruonline.com

http://www.mjpruonline.com

7. (a) Show that :

$$\bar{\nabla} r^n = n r^{n-2} \bar{r}$$

(b) Show that the vectors $\bar{a} = 3\hat{i} - 2\hat{j} + \hat{k}$, $\bar{b} = \hat{i} - 3\hat{j} + 5\hat{k}$ and $\bar{c} = 2\hat{i} + \hat{j} - 4\hat{k}$ form a right angled triangle.

8. (a) Find the centre, vertices, foci, eccentricity and letus rectum of the hyperbola :

$$9x^2 - 16y^2 - 18x - 64y - 199 = 0$$

(b) One extremity of focal chord of the parabola $y^2 = 16x$ is the point (1, 4), find the co-ordinates of the other extremity.

http://www.mjpruonline.com

http://www.mjpruonline.com

http://www.mjpruonline.com
Whatsapp @ 9300930012
Your old paper & get 10/-
पुराने पेपर्स भेजे और 10 रुपये पायें,
Paytm or Google Pay से

http://www.mjpruonline.com